

Unit 1 : Basic of Data Warehousing

Mark - 18

List & Explain Benefits of Data Warehousing or Needs

Benefits of Data Warehousing :-

1. Integrated Data from Multiple Sources
2. Support for Better Decision Making
3. Historical Data Analysis
4. Improved Data quality & Performance
5. Faster Query Performance
6. Improved Data Consistency
7. Separation of Analytical & Transactional Workload
8. Business Intelligence & Reporting

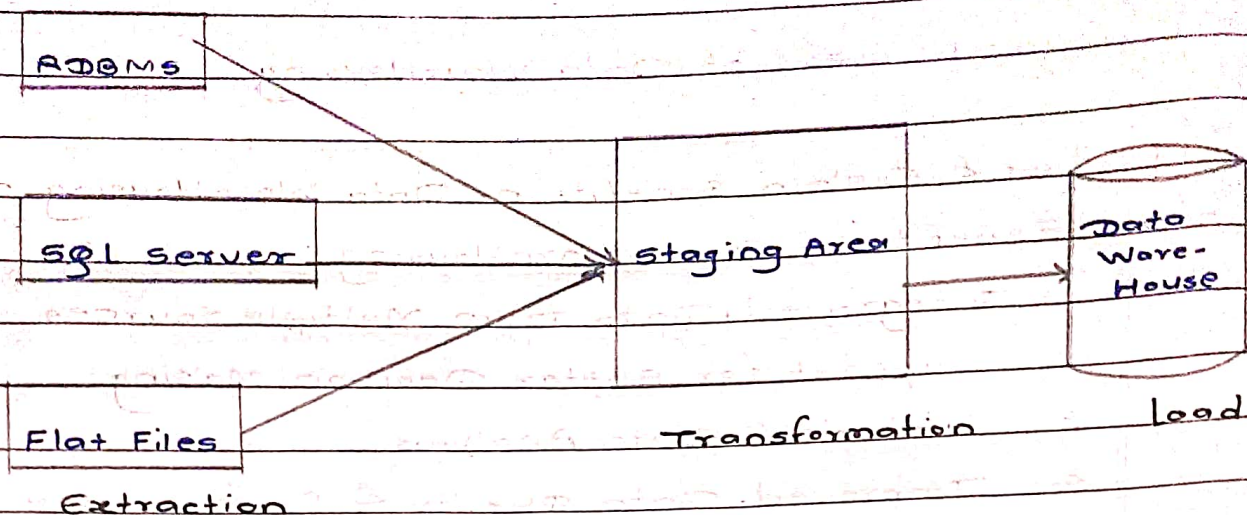
• Needs of Data Warehousing :

- Organization Generate Huge Amount of Data from Different System such as sales, Marketing & Operations
- The Data is often Scattered, inconsistent & Difficult to Analyze.
- Data Warehouse is Needed to Overcome These problem & Support Better Decision Making

Q3] Explain ETL process with Neat Diagram

→ ETL (Extract, Transformed, load) is a Data Integration process Used in Data Warehousing & Data Analytics.

- It Collects Data from Different sources, Converts it into a Useful format & stores it in a Data Warehouse for Reporting & Decision Making



1. Extract :

Extraction is The process of Collecting Raw Data from Multiple Data sources such as Databases (MySQL, Oracle, SQL server), Flat files (CSV, Excel, XML, JSON), API's, ERP, CRM, Sensors, log & IoT Devices.

2. Transformation :

Transformation Converts Raw Data into a clean, standardized & Business-ready format.

3. Load :

Loading is The process of storing Transformed Data into The Data Warehouse.

Q.4] Differentiate Both Operational Database & Data Warehouse.

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Aspect	Operational D.B.	Data Warehouse
1. Purpose	It is Used for Daily Transaction	It is Used for Analysis & Reporting

2. Data

The Data is Current Data

- Historical Data

3. Updates

- frequent

- Periodic

4. Users

Clerks, Application

Manager, Analysts

Q5] Explain with suitable ex. Data Warehouse with their four characteristics.

→ Data warehouse: Data Warehouse is Central Repository for Integrated Consistent Historical Data from Multiple sources.

- It act as a Central library

- It stores & organized large Historical Data sets for easy Access.

- Data Warehouse is subject oriented, Integrated, Time Variant (Historical) & Non-Volatile Collection of Data from Multiple sources to support Decision Support, Making Analysis & Reporting.

• Characteristics:

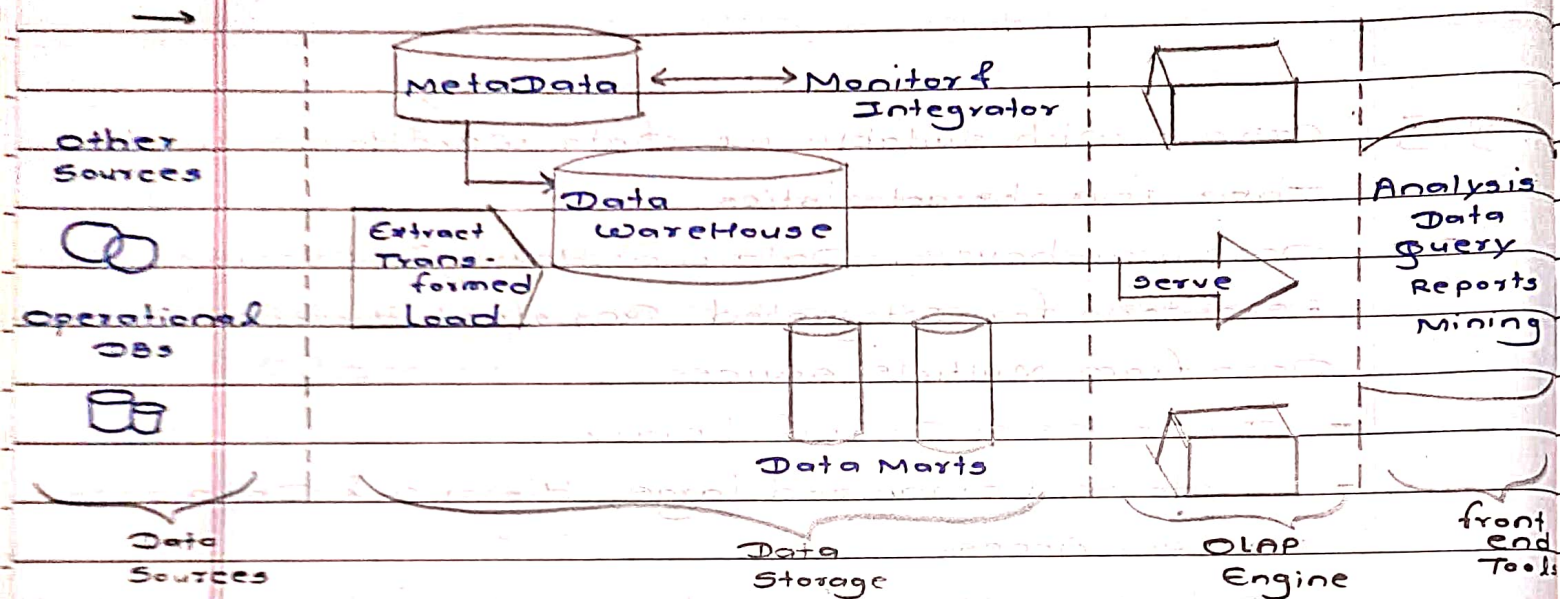
1. Subject oriented: Data is organized Around Major subject such as sales, Marketing, Customer product rather Than Business process.

2. Integrated: Data from Different sources is clean & stored in Consistent format.

3. Time Variant (Historical) : Historical Data is Maintained over long Period, enabling Trend Analysis.

4. Non-Volatile : Data is stable once it is stored Non frequently Updated or Deleted

Q.6] Explain Three Tier Architecture of Data Warehouse With Diagram.



- Three Tier Architecture is widely used for its clear structure Dividing into Data processing into Three Layers for efficient Access & Management.

• layers :

1. Bottom Tier (Data sources & Data storage) :

It is foundation of Data warehouse Responsible for Collecting , processing & storing Data from Multiple Sources.

- Key Component of Bottom Tier is Data sources & Data storage

2. Middle Tier: It Host The OLAP server which process Complex Analytical queries

- It Act as Bridge Between Data storage Layer & User Interface

- It ensures Data is quickly Retrieve, Aggregated & Ready for Reporting

• OLAP Server Model:

1. ROLAP: Relational OLAP - Used Relational D.B to stored & Manage Warehouse Data. It is Ideal to Handling Large Data because it operate On Relational D.B Directly.

2. MOLAP: Multidimensional OLAP - stored Data in Multidimensional queue.

The storage & Retrieval process are Highly efficient Making suitable for Complex Analytical queries

3. HOLAP: Hybrid OLAP:- It is Combination of Relational & MOLAP Analytical processing

- It is The Ideal Option for seamless functional flow Across D.B.

3. Top Tier (front end Tools): It is front-end client layer which is essential for interacting with Data stored & process in lower layer

- This layer include Variety of Business Intelligence (BI) Tools, & Techniques for Reporting Analysis & Decision Making

Q.3] Differentiate Between Data Warehouse & Data Mart.

Q.3] Explain Metadata Repository:-



Basis	Data Warehouse	Data Mart
1. Definition	A Data warehouse is a centralized Repository that stores Data from The entire organization.	A Data Mart is a Subset of a Data warehouse Designed for a specific Dept.
2. Scope	Covers The whole organization.	Covers one Department (Sales, HR, finance)
3. Size	Very large in size.	Smaller in size.
4. Data sources	Collects Data from Multiple sources.	Usually Collects Data from a single Source or Warehouse.
5. Purpose	Used for enterprise level Decision Making.	Used for Dept-level Decision Making.
6. Complexity & Cost	More Complex & High Cost.	Low Cost & Less Complex.
7. Ex	Company wide BI system.	Sales Data Mart HR Data Mart.

Q.8) Explain Data warehouse Models

→ • Data warehouse Models :

1. Enterprise Data warehouse (EDW) :-

A Central, Integrated Repository Consolidating Historical Data from Across an entire organizations (Sales, finance, operations etc) into a single source of Truth for Company-wide Business Intelligence, Complex Analytics & informed Decision Making supporting

- It serves The whole organizations, EDWs store Clean, structured Data.

• How It works ?

1. Data sources : Data is Extracted from Operational D.B, SaaS platform, spreadsheets & more.

2. ETL / ELT process :- Data is Extracted, Transformed (cleaned, formatted) & loaded into The warehouse.

3. Central Repository (EDW) :- stores Integrated Historical Data in a structured format.

4. Analytics & Reporting : Users Access The EDW Via BI Tools for Dashboards, Custom Reports & Analysis.

2. Data Mart : It is a specialized subset of Data warehouse focused on a specific functional Area or Department within an organization.

It provides a simplified & Target View of Data, Addressing specific Reporting & Analytical needs.

- Data Mart are smaller in scale & scope, typically holding relevant data.

• Types :

1. Independent Data Mart : It is created & maintained separately from the data warehouse. It is created to satisfy the particular needs of specific business unit.

2. Dependent Data Mart : It is generated right out of data warehouse. It takes some data from the data warehouse & arranges it to meet the needs of industry.

3. Hybrid Data Mart : Both dependent & independent data mart components can be found in hybrid data marts.

3. Virtual Warehouse :-

- It provides a logical view of data from multiple sources without physically storing the data in central location.

• features :

1. Data Integration : Combines data from various sources without physical movement.

2. Real Time Access : provides up to date info. directly from source systems.

3. Reduced storage costs : Minimizes the need for additional storage infrastructure.

4. flexibility : easily adapts to changes in source system.

Q.9] state The Usage of Data warehousing

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1. Decision Support : Helps Managements make better & faster Decision Using Accurate & Historical Data
 2. Business Intelligence & Reporting : Used to Generate Reports, Dashboards & Analytical Summaries
 3. Trend & Pattern Analysis : Analyzes past Data to Identify Trends, patterns & future opportunities
 4. Data Integration : Combines Data from Multiple Sources into a single Centralized Systems.
 5. Performance Analysis : Evaluates Business performance such as sales, profit & productivity.
 6. Data Consistency & quality : Provides clean, Reliable & Consistent Data for Analysis.

Q.10) Explain Major Tasks in Data preprocessing

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1. Data Cleaning :
Removes error, noise & inconsistencies from Data
ex : Handling Missing values, Removing Duplicates, Correcting wrong Data.
 2. Data Integration :
Combines Data from Multiple source into a single Dataset.
ex : Merging Data from Databases, files or Data Warehouses.
 3. Data Transformation : Converts Data into suitable for Analysis or Mining
ex : Normalization, aggregation, smoothing & encoding.

4. Data Reduction: Reduces Data Size while preserving important Information.
ex: Dimensionally Reduction, Sampling, Data Compression.